Assignment Beta Question 5

Research Paper Summary

Paper: *A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications: Design, Implementation, and Assessment*
Authors: Pinar Ersoy and Mustafa Ersahin
Published in: *IEEE Access*, 2025

Overview and Research Objective

Ersoy and Ersahin investigate how retrieval-augmented generation (RAG) can enhance the reliability of large language model (LLM) applications in knowledge-heavy environments. While standard LLMs can generate coherent responses, their internal information may be outdated, and they can create incorrect factual statements. These vulnerabilities pose significant risks in sectors like finance and e-commerce, where an inaccurate monetary amount, product policy, or regulatory limit can lead to financial losses. RAG mitigates this issue by retrieving pertinent external text during inference and providing it as evidence to the generator.

The paper analyzes four primary RAG architectures: sparse, dense, hybrid, and fusion-based retrieval. Its aim is not to propose a new retriever, but to present a reproducible and deployment-focused approach for determining which architecture is most suitable for a specific use case. The authors evaluate the systems based on factual grounding, relevance, retrieval quality, latency, and auditability, and compare all pipelines against a zero-shot baseline that utilizes the same generator but lacks retrieved context.

Architectures Compared

Architecture	Retrieval approach	Main strengths	Main limitations
Sparse RAG	BM25 or another lexical index retrieves passages with exact or near-exact term overlap.	Fast, resource-efficient, transparent, and easy to audit.	Misses relevant passages when users paraphrase or use synonyms.
Dense RAG	Queries and passages are converted into neural embeddings to find	Strong semantic recall for paraphrased and	Costlier indexing and similarity scores that

Architecture	Retrieval approach	Main strengths	Main limitations
	semantically similar evidence.	multilingual questions.	are difficult to explain.
Hybrid RAG	Sparse and dense retrieval run together; ranked results are merged.	Balances exact-match precision with semantic coverage.	Requires two indexes and more maintenance.
Fusion RAG	Multiple searches, sub-queries, or indexes are combined before or during generation.	Highest factual coverage, faithfulness, and relevance for difficult queries.	Adds latency, cost, and complexity; attribution becomes less clear.

Methodology and Evaluation Design

The authors focus on finance and e-commerce because these fields necessitate answers that are numerically accurate, up-to-date, and verifiable. Their testing dataset comprises 500 questions, organized by domain and level of difficulty. Seventy-five percent of these questions can be answered based on the collected evidence, while twenty-five percent are intentionally unanswerable, which ensures that a system does not seem reliable simply by choosing not to respond. The evidence comes from primary and verifiable sources, such as regulatory filings and official product and policy pages.

The generator remains constant across all experimental conditions so that any variations can be linked to retrieval rather than the language model. The study uses RAGAS-style metrics (faithfulness, answer relevance, context precision, and context recall) along with latency and **auditability**, which is defined as the percentage of answer claims that are backed by valid and adequately recent citations. The findings include 95% bootstrap confidence intervals and paired statistical analyses rather than just single point estimates.

Major Findings

All retrieval-based systems demonstrate enhanced factual accuracy compared to zero-shot generation. The ungrounded GPT-4o baseline misrepresented monetary amounts or outdated regulatory limits in about one-third of instances, while each RAG pipeline lowered such inaccuracies to single-digit percentages.

Sparse retrieval provides the highest context precision and auditability due to its checkable lexical scoring, making it appealing for highly regulated settings with stringent latency

requirements. Nevertheless, it yielded lower recall when questions used paraphrases, different vocabulary, or multilingual expressions.

Dense retrieval enhanced semantic reach by identifying conceptually related information, even with minimal word overlap. However, it comes with reduced interpretability, higher indexing expenses, and a less robust audit trail.

Hybrid retrieval provides the best overall balance. By integrating BM25's precision with the semantic depth of dense retrieval, it ensures high accuracy and relevant answers while maintaining a reasonable latency for interactive applications.

Fusion systems delivered the highest levels of accuracy, answer relevance, and recall, and are particularly beneficial for ambiguous, intricate, or cross-domain queries. However, their benefits come with increased latency, higher prompt costs, and reduced clarity regarding the connection between source passages and the claims produced.

Limitations and Validity Considerations

The findings are most robust in finance and e-commerce, and results may vary in fields with different document formats, freshness needs, or terminology. The authors also mention that automatic RAGAS-type metrics serve as substitutes for human evaluation, which they mitigate by examining a stratified sample and checking claim-level citation consistency. The study also highlights that the quality and timeliness of sources are as important as the retrieval architecture itself.

Conclusion

Sparse RAG emerges as the clearest and most efficient option when auditability and precise terminology are the main concerns, while dense RAG is beneficial when semantic or multilingual coverage matters most. Hybrid RAG is the suggested standard for numerous interactive applications since it offers the best overall mix of precision, recall, faithfulness, and response time. Fusion RAG is ideally suited for scenarios where comprehensive factual coverage is essential and longer latency can be tolerated, such as batch processing or critical analytical work. The authors also provide a reproducible evaluation framework, and future research should explore adaptive indexing, retrieval-aware decoding, and human assessment across more domains.

Reference

Ersoy, P., & Ersahin, M. (2025). *A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications: Design, Implementation, and Assessment*. *IEEE Access, 13*, 194185-194196. https://doi.org/10.1109/ACCESS.2025.3632404